

Soumyajyoti Dutta

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Empirical LLM Research | Post-Training & Evaluation | AI for Cyberdefense & Adversarial ML

EXPERTISE

Core Competencies:

- **LLM post-training:** supervised fine-tuning and curriculum-staged training to align foundation models (135M–~9B, incl. sparse MoE) to intended behavior — e.g. teaching an agent to clarify rather than hallucinate on ambiguous intent.
- **Evaluation & Diagnosis:** task-specific metrics (lexical + syntactic + SMT-based semantic), LLM-as-judge and user-simulation harnesses deployed as training-time quality gates; stage-wise ablations that trace *where* a method's gains stop and *why*.
- **AI × security:** LLM systems for automated cyberdefense (threat-intel rule generation, IOC/TTP extraction), adversarial ML (built a competition's top-ranked adversarial attack suite), and ML malware detection at 2M+-sample scale.
- **Efficient training & inference:** single-GPU & multi-node SLURM training with FFT and PEFT (rank-stabilized LoRA), MoE, grammar-derived hybrid tokenization, and vLLM inference with multi-turn KV reuse.
- **Research → public output:** USENIX Security '27 Cycle-1 submission (second targeting Cycle 2), ACSAC 2026 acceptance with an open-source PyPI release, Springer chapter (2022), and a deployed full-stack applied-LLM platform.

Technical Stack:

Python (PyTorch, HuggingFace Transformers, TRL, TensorFlow, NumPy, Pandas, scikit-learn, Matplotlib), SQL/SQLite, DeepSpeed, FSDP, DDP/NCCL, CUDA, SLURM, vLLM, Z3/SMT, AWS, Git, CI/CD, FastAPI/Flask/Django, React, C++, Rust

PROFESSIONAL EXPERIENCE

Texas A&M University — Botacin's Lab

Research Assistant / Machine Learning Engineer (Applied AI)

College Station, TX

Spring 2024 – Present

- Built end-to-end LLM post-training pipelines: data curation, SFT / curriculum training, strategic evaluation, and deployment across ~a dozen model families (135M–~9B) on multi-node SLURM/GPU clusters.
- **YaraGen**, sponsored by **Laboratory For Physical Sciences, UMD & Booz Allen Hamilton** — grammar-oriented minimal LLMs for automated YARA rule generation (**submitted, USENIX Security '27, Cycle 1**):
 - * Lifted a **139M-parameter BART** to **92% semantic correctness** (93.7% in-window) on 22.7K held-out real-world rules — matching a fully fine-tuned 8B model (93.8%) at **57× fewer parameters** and 17–20× the throughput, and leading few-shot GPT-5.4 by ~10 pp.
 - * Ran **150+ post-training runs** across BART, CodeT5p, T5Gemma, SmolLM2, LFM2.5 (230M–1.2B), Gemma 3 (270M–1B), Llama 3.1 8B / 3.2 1B, and Ministral 3 8B, spanning FFT, LoRA/QLoRA, and seq2seq vs. causal ablations.
 - * Designed the **YaraAST hybrid tokenizer** — grammar-derived atomic tokens for rule structure over base subwords for IOC content — yielding +4 pp accuracy, 66% lower training loss, and 11% per-rule context reduction, model-agnostic via HuggingFace.
 - * Co-designed a **55M BiLSTM NER Scooper** that masks IOCs behind placeholders (99.4% precision / 99.6% recall), adding up to +10 pp correctness and decoupling IOC size from the context window.
 - * Assembled three corpora (305K crawled in-the-wild rules; 15M synthetic from 459 curated templates; 15M grammar-fuzzed) and built *Brownie & Puff* composite evaluators (BLEU + parse/compile + **Z3 SMT semantic equivalence**) as training-time quality gates for all runs.
 - * Downstream on EMBER malware clusters, generated rules reach **96±4% TP at 0.8% FP** (99.7% generation success), beating AutoYara; shipped the full framework, agentic web chatbot, and models as open-source artifacts.
- **IOCScooper++** — IOC/TTP extraction from real threat reports (**targeting USENIX Security '27, Cycle 2**):
 - * Trained **ScooperNER**, a 50.6M-parameter extractor (MLM pretraining → fine-tuning, 36 IOC types): **0.894 micro / 0.925 macro** span-exact F1 on held-out vendor and government reports, beating fine-tuned encoder baselines at 3–7× its size, with 0.950 F1 on explicit MITRE ATT&CK IDs.
 - * End to end on 47 held-out reports, extraction feeds frozen YaraGen at 0.86/0.88 precision/recall with **100% rule validity** at **0.97 IOC coverage** across 344 generated rules.

- * Extending extraction to grounded (entity, action, IOC) tuples lifted into ATT&CK techniques via procedure-example retrieval, open-set over all 637 techniques, with a full-report co-annotated benchmark of real malware analysis reports.

- **AutoPYara** — shipped an open-source rule-generation package (PyPI; **ACSAC 2026, accepted**): +14 pp coverage, 7.1 pp lower variance, and +10% threat-hunting accuracy over AutoYara (prior state of the art).

Cognizant Technology Solutions

Kolkata, India

Junior Software Engineer — Client-Embedded Consulting (Client: TJX Companies)

Aug 2021 — Aug 2022

- Client-embedded consultant for TJX Companies (Fortune 500): shipped a logistics-automation platform; optimized Python/REST services (latency, concurrency, CI/CD) and led cross-functional code reviews.

Software Engineering Intern — converted to full-time offer

Feb 2021 — Jun 2021

- Built web features for a client logistics platform; resolved support tickets via ServiceNow triage and Splunk tracing.

SELECTED PROJECTS

Ambiguity-Aware Research Agent | *Co-authored (manuscript in prep.) — led model development & training* 2026

- Aligned a deep-research agent to pause and ask a clarifying question on underspecified intent — rather than confidently answer the wrong question — via a four-stage SFT curriculum on Gemma-4-E4B (sparse MoE, ~9B/4B), single 80GB GPU + vLLM with multi-turn KV reuse.
- Built a three-axis evaluation (detection, clarification quality, answer quality) with GPT-4o user-simulation and LLM-as-judge; lifted pause-detection F1 from 0.168 to 0.573, and reported the honest negative: clarification-conditioned answers still trail the no-pause baseline, isolating where SFT alone is insufficient.

Depixelation Recoverability Frontier | *Personal vision-security research — in progress*

2026

- Designing an open-source benchmark mapping when pixelated text is recoverable — a **360-cell** difficulty grid (block size × font-blindness × corruption × charset entropy) with a **10-seed** statistical protocol — reframing depixelation from image restoration to constrained search with an exact render-and-verify check.
- Scoped retrieval-augmented recognition with calibrated abstention and curriculum-training comparisons on 4×H100 — with a defense-first responsible-use framing.

Sturdy Fishstick — Full-Stack Applied-LLM Platform | *FastAPI, React, local LLMs via Ollama*

2025

- Deployed a full-stack applied-LLM platform that has scored ~750 job listings from 10+ ingestion sources, using a categorical-scoring design that turns 1–2B local models into reliable deterministic judges and FTS5-grounded RAG that prevents hallucinated statistics by construction.

Multimodal & Vision Evaluation | *ECEN 642 — Image Processing & Computer Vision*

2024

- Evaluated a DeepSeek vision-language model against human-annotated ground truth; earlier: carotid-artery ultrasound CNN classification (Springer chapter, 2022) and WildFire spatiotemporal prediction.

ML Malware Detection & Adversarial Attack Suite | *1st Place; still #1 after five course offerings*

2023

- Built an end-to-end Windows PE malware classifier (EMBER/BODMAS/Benign-NET, 2M+ samples; LightGBM/XGBoost) — 96.2% accuracy, 0.97 F1 — served from a self-contained Docker HTTP endpoint; designed the competition's best-performing adversarial attack suite against peer defenses.

EDUCATION

Ph.D. in Computer Science — Texas A&M University

Spring 2024 – Present

- *Research focus:* LLM post-training, evaluation, and interpretability for automated cyberdefense.

M.S. in Computer Engineering — Texas A&M University

Fall 2022 – Fall 2023

- *Coursework:* NLP, Deep Learning, Image Processing & Computer Vision, ML Cyberdefenses, ML Theory, Data Mining, Randomized Algorithms.

B.Tech. in Electronics & Communication Engineering — SRM University

Fall 2017 – Spring 2021

PUBLICATIONS & MANUSCRIPTS

YaraGen — minimal grammar-oriented LLMs for YARA rule generation

USENIX Security '27, submitted (Cycle 1)

IOCScooper++ — grounded IOC/TTP extraction from threat reports

USENIX Security '27, in prep. (Cycle 2)

AutoPYara — automated rule generation; open-sourced on PyPI

ACSAC 2026, accepted

Proactive Deep Research Agents — ambiguity-aware clarification (co-authored)

2026, in preparation

Carotid-Artery Ultrasound Classification — CNN-based medical imaging

Springer book chapter, 2022

TEACHING & OUTREACH

Teaching Assistant, CSCE 313 — Introduction to Computer Systems, Texas A&M

Fall 2026

- Lead systems-programming labs: processes, IPC, multithreading/synchronization, and socket programming.

Teaching Assistant, CSCE 439/704 — Data Analytics for Cybersecurity, Texas A&M

Spring 2026

- Taught clustering, supervised ML, and anomaly detection to 37 students; mentored 9 team projects.

Partner & Presenter, TAMU CS Day — interactive AI/cybersecurity demos for high-school students

2024, 2025